

Regularized Multivariate Two-way Functional Principal Component Analysis

Mobina Pourmoshir

`mobina.pourmoshir@marquette.edu`

Department of Mathematical and Statistical Sciences
Marquette University



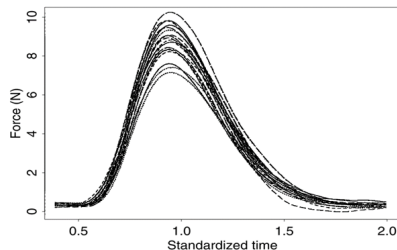
Outline

- 1 Introduction & Background
- 2 A SVD Approach for Regularized Multivariate FPCA
- 3 Two-way Regularized Multivariate FPCA
- 4 Conclusion & Future Work

Introduction

Functional Data Analysis:

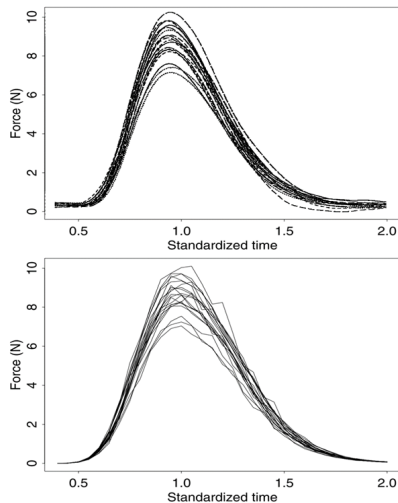
- Functional data are observations that vary continuously over a domain (like time, space, or wavelength) and are often visualized as curves, trajectories, or functions rather than isolated points.



Introduction

Functional Data Analysis:

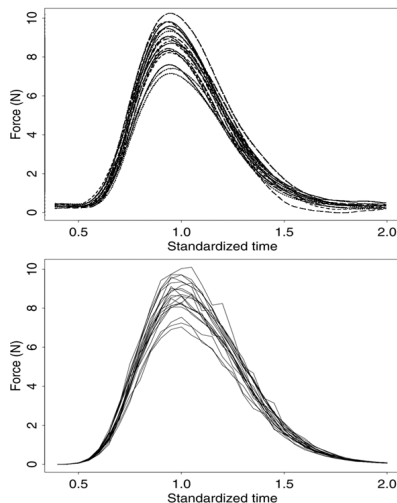
- Functional data are observations that vary continuously over a domain (like time, space, or wavelength) and are often visualized as curves, trajectories, or functions rather than isolated points.
- These data are often recorded at discrete time points or grid locations, even though they originate from continuous processes in areas like engineering, finance, environmental science, and healthcare.



Introduction

Functional Data Analysis:

- Functional data are observations that vary continuously over a domain (like time, space, or wavelength) and are often visualized as curves, trajectories, or functions rather than isolated points.
- These data are often recorded at discrete time points or grid locations, even though they originate from continuous processes in areas like engineering, finance, environmental science, and healthcare.
- Functional Data Analysis (FDA) is a statistical framework that treats these observations as realizations of smooth underlying functions, allowing for more accurate modeling and interpretation of continuous processes.



Functional Principal Component Analysis

- Functional PCA (FPCA): Extension of classical PCA to functional data; finds orthogonal functions capturing main variation [Ramsay and Silverman, 2005].

Functional Principal Component Analysis

- Functional PCA (FPCA): Extension of classical PCA to functional data; finds orthogonal functions capturing main variation [Ramsay and Silverman, 2005].
- Purpose: Dimension reduction and uncovering hidden patterns.

Functional Principal Component Analysis

- Functional PCA (FPCA): Extension of classical PCA to functional data; finds orthogonal functions capturing main variation [Ramsay and Silverman, 2005].
- Purpose: Dimension reduction and uncovering hidden patterns.
- Advancements in FPCA:

Functional Principal Component Analysis

- Functional PCA (FPCA): Extension of classical PCA to functional data; finds orthogonal functions capturing main variation [Ramsay and Silverman, 2005].
- Purpose: Dimension reduction and uncovering hidden patterns.
- Advancements in FPCA:
 - Smoothed FPCA: Adds roughness penalties for smoothness [Silverman, 1996, Huang et al., 2008].

Functional Principal Component Analysis

- Functional PCA (FPCA): Extension of classical PCA to functional data; finds orthogonal functions capturing main variation [Ramsay and Silverman, 2005].
- Purpose: Dimension reduction and uncovering hidden patterns.
- Advancements in FPCA:
 - Smoothed FPCA: Adds roughness penalties for smoothness [Silverman, 1996, Huang et al., 2008].
 - Sparse FPCA: Enforces sparsity for interpretability [Shen and Huang, 2008, Nie and Cao, 2020].

Functional Principal Component Analysis

- Functional PCA (FPCA): Extension of classical PCA to functional data; finds orthogonal functions capturing main variation [Ramsay and Silverman, 2005].
- Purpose: Dimension reduction and uncovering hidden patterns.
- Advancements in FPCA:
 - Smoothed FPCA: Adds roughness penalties for smoothness [Silverman, 1996, Huang et al., 2008].
 - Sparse FPCA: Enforces sparsity for interpretability [Shen and Huang, 2008, Nie and Cao, 2020].
 - Multivariate FPCA (MFPCA): Extends FPCA to multivariate functions [Silverman, 1996, Happ and Greven, 2018].

Functional Principal Component Analysis

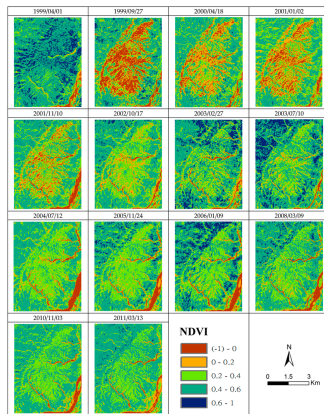
- Functional PCA (FPCA): Extension of classical PCA to functional data; finds orthogonal functions capturing main variation [Ramsay and Silverman, 2005].
- Purpose: Dimension reduction and uncovering hidden patterns.
- Advancements in FPCA:
 - Smoothed FPCA: Adds roughness penalties for smoothness [Silverman, 1996, Huang et al., 2008].
 - Sparse FPCA: Enforces sparsity for interpretability [Shen and Huang, 2008, Nie and Cao, 2020].
 - Multivariate FPCA (MFPCA): Extends FPCA to multivariate functions [Silverman, 1996, Happ and Greven, 2018].
 - Regularized MFPCA: Penalties improve estimation & interpretability [Hagbabin et al., 2023].

Functional Principal Component Analysis

- Functional PCA (FPCA): Extension of classical PCA to functional data; finds orthogonal functions capturing main variation [Ramsay and Silverman, 2005].
- Purpose: Dimension reduction and uncovering hidden patterns.
- Advancements in FPCA:
 - Smoothed FPCA: Adds roughness penalties for smoothness [Silverman, 1996, Huang et al., 2008].
 - Sparse FPCA: Enforces sparsity for interpretability [Shen and Huang, 2008, Nie and Cao, 2020].
 - Multivariate FPCA (MFPCA): Extends FPCA to multivariate functions [Silverman, 1996, Happ and Greven, 2018].
 - Regularized MFPCA: Penalties improve estimation & interpretability [Hagbini et al., 2023].
- Impact: More adaptable, robust, and applicable across diverse scientific and business problems.

Two- way Functional Principal Component Analysis

- Two-way functional data: Observations vary along two domains (e.g., time $\times$ space, time $\times$ frequency), with applications in climate, neuroscience, finance, public health, and marketing.



Two- way Functional Principal Component Analysis

- Two-way functional data: Observations vary along two domains (e.g., time $\times$ space, time $\times$ frequency), with applications in climate, neuroscience, finance, public health, and marketing.
- Extension of FPCA: Huang [Jianhua Z. Huang and Buja, 2009] applied regularization to both left and right singular vectors in SVD.

Two- way Functional Principal Component Analysis

- Two-way functional data: Observations vary along two domains (e.g., time $\times$ space, time $\times$ frequency), with applications in climate, neuroscience, finance, public health, and marketing.
- Extension of FPCA: Huang [Jianhua Z. Huang and Buja, 2009] applied regularization to both left and right singular vectors in SVD.
- Practical challenges:

Two- way Functional Principal Component Analysis

- Two-way functional data: Observations vary along two domains (e.g., time $\times$ space, time $\times$ frequency), with applications in climate, neuroscience, finance, public health, and marketing.
- Extension of FPCA: Huang [Jianhua Z. Huang and Buja, 2009] applied regularization to both left and right singular vectors in SVD.
- Practical challenges:
 - Data observed on discrete grids (minutes, hours, days).

Two- way Functional Principal Component Analysis

- Two-way functional data: Observations vary along two domains (e.g., time $\times$ space, time $\times$ frequency), with applications in climate, neuroscience, finance, public health, and marketing.
- Extension of FPCA: Huang [Jianhua Z. Huang and Buja, 2009] applied regularization to both left and right singular vectors in SVD.
- Practical challenges:
 - Data observed on discrete grids (minutes, hours, days).
 - Issues: measurement noise, irregular sampling, missing data, loss of smoothness.

Two- way Functional Principal Component Analysis

- Two-way functional data: Observations vary along two domains (e.g., time $\times$ space, time $\times$ frequency), with applications in climate, neuroscience, finance, public health, and marketing.
- Extension of FPCA: Huang [Jianhua Z. Huang and Buja, 2009] applied regularization to both left and right singular vectors in SVD.
- Practical challenges:
 - Data observed on discrete grids (minutes, hours, days).
 - Issues: measurement noise, irregular sampling, missing data, loss of smoothness.
- Proposed framework:

Two- way Functional Principal Component Analysis

- Two-way functional data: Observations vary along two domains (e.g., time $\times$ space, time $\times$ frequency), with applications in climate, neuroscience, finance, public health, and marketing.
- Extension of FPCA: Huang [Jianhua Z. Huang and Buja, 2009] applied regularization to both left and right singular vectors in SVD.
- Practical challenges:
 - Data observed on discrete grids (minutes, hours, days).
 - Issues: measurement noise, irregular sampling, missing data, loss of smoothness.
- Proposed framework:
 - Unified FPCA for two-way multivariate functional data.

Two- way Functional Principal Component Analysis

- Two-way functional data: Observations vary along two domains (e.g., time $\times$ space, time $\times$ frequency), with applications in climate, neuroscience, finance, public health, and marketing.
- Extension of FPCA: Huang [Jianhua Z. Huang and Buja, 2009] applied regularization to both left and right singular vectors in SVD.
- Practical challenges:
 - Data observed on discrete grids (minutes, hours, days).
 - Issues: measurement noise, irregular sampling, missing data, loss of smoothness.
- Proposed framework:
 - Unified FPCA for two-way multivariate functional data.
 - Smoothness penalties preserve functional structure.

Two- way Functional Principal Component Analysis

- Two-way functional data: Observations vary along two domains (e.g., time $\times$ space, time $\times$ frequency), with applications in climate, neuroscience, finance, public health, and marketing.
- Extension of FPCA: Huang [Jianhua Z. Huang and Buja, 2009] applied regularization to both left and right singular vectors in SVD.
- Practical challenges:
 - Data observed on discrete grids (minutes, hours, days).
 - Issues: measurement noise, irregular sampling, missing data, loss of smoothness.
- Proposed framework:
 - Unified FPCA for two-way multivariate functional data.
 - Smoothness penalties preserve functional structure.
 - Sparsity penalties enhance interpretability.

Two- way Functional Principal Component Analysis

- Two-way functional data: Observations vary along two domains (e.g., time $\times$ space, time $\times$ frequency), with applications in climate, neuroscience, finance, public health, and marketing.
- Extension of FPCA: Huang [Jianhua Z. Huang and Buja, 2009] applied regularization to both left and right singular vectors in SVD.
- Practical challenges:
 - Data observed on discrete grids (minutes, hours, days).
 - Issues: measurement noise, irregular sampling, missing data, loss of smoothness.
- Proposed framework:
 - Unified FPCA for two-way multivariate functional data.
 - Smoothness penalties preserve functional structure.
 - Sparsity penalties enhance interpretability.
 - Effective for dimension reduction in complex datasets.

Mathematical Foundation of FPCA

- **Goal:** Identify functional directions that maximize variance (low-rank approximation of functional data). For functional data $X \in \mathbb{R}^{n \times m}$ contains the discretized functional observations (rows correspond to subjects, columns to grid points), $v \in \mathbb{R}^m$ represents the estimated principal component (function), and $u \in \mathbb{R}^n$ denotes the associated principal component scores.

Mathematical Foundation of FPCA

- **Goal:** Identify functional directions that maximize variance (low-rank approximation of functional data). For functional data $X \in \mathbb{R}^{n \times m}$ contains the discretized functional observations (rows correspond to subjects, columns to grid points), $v \in \mathbb{R}^m$ represents the estimated principal component (function), and $u \in \mathbb{R}^n$ denotes the associated principal component scores.
- **Reconstruction problem:**

$$\min_{u,v} \|X - uv^\top\|_F^2 = \text{tr}\{(X - uv^\top)(X - uv^\top)^\top\},$$

Mathematical Foundation of FPCA

- **Goal:** Identify functional directions that maximize variance (low-rank approximation of functional data). For functional data $X \in \mathbb{R}^{n \times m}$ contains the discretized functional observations (rows correspond to subjects, columns to grid points), $v \in \mathbb{R}^m$ represents the estimated principal component (function), and $u \in \mathbb{R}^n$ denotes the associated principal component scores.
- **Reconstruction problem:**

$$\min_{u,v} \|X - uv^\top\|_F^2 = \text{tr}\{(X - uv^\top)(X - uv^\top)^\top\},$$

- **Optimization steps:**

$$\text{Fix } v : u = \frac{Xv}{v^\top v} \quad \text{and} \quad \text{Fix } u : v = \frac{X^\top u}{u^\top u}$$

Extensions of FPCA via Regularization

- Goal: Balance **variance explanation**, **smoothness**, and **interpretability**.

Extensions of FPCA via Regularization

- Goal: Balance **variance explanation**, **smoothness**, and **interpretability**.
- Reformulate FPCA as a **penalized low-rank approximation** problem:

$$\min_{u,v} \|X - uv^{\top}\|_F^2 + \mathcal{P}(u, v)$$

Extensions of FPCA via Regularization

- Goal: Balance **variance explanation**, **smoothness**, and **interpretability**.
- Reformulate FPCA as a **penalized low-rank approximation** problem:

$$\min_{u,v} \|X - uv^{\top}\|_F^2 + \mathcal{P}(u, v)$$

- Two directions:

Extensions of FPCA via Regularization

- Goal: Balance **variance explanation**, **smoothness**, and **interpretability**.
- Reformulate FPCA as a **penalized low-rank approximation** problem:

$$\min_{u,v} \|X - uv^{\top}\|_F^2 + \mathcal{P}(u, v)$$

- Two directions:
 - **Smooth FPCA**: adds roughness penalty on functions.

Extensions of FPCA via Regularization

- Goal: Balance **variance explanation**, **smoothness**, and **interpretability**.
- Reformulate FPCA as a **penalized low-rank approximation** problem:

$$\min_{u,v} \|X - uv^{\top}\|_F^2 + \mathcal{P}(u, v)$$

- Two directions:
 - **Smooth FPCA**: adds roughness penalty on functions.
 - **Sparse FPCA**: adds sparsity penalty on loadings.

Extensions of FPCA via Regularization

- Goal: Balance **variance explanation**, **smoothness**, and **interpretability**.
- Reformulate FPCA as a **penalized low-rank approximation** problem:

$$\min_{u,v} \|X - uv^{\top}\|_F^2 + \mathcal{P}(u, v)$$

- Two directions:
 - **Smooth FPCA**: adds roughness penalty on functions.
 - **Sparse FPCA**: adds sparsity penalty on loadings.
- Algorithms: Based on iterative **power method** and **thresholding** updates.

Smooth Functional PCA

- Problem setup [Huang et al., 2008]:

$$\min_{u,v} \|X - uv^T\|_F^2 + \alpha u^T u v^T \Omega v$$

Smooth Functional PCA

- Problem setup [Huang et al., 2008]:

$$\min_{u,v} \|X - uv^T\|_F^2 + \alpha u^T u v^T \Omega v$$

- $X \in \mathbb{R}^{n \times p}$: discretized functional data.

Smooth Functional PCA

- Problem setup [Huang et al., 2008]:

$$\min_{u,v} \|X - uv^T\|_F^2 + \alpha u^T u v^T \Omega v$$

- $X \in \mathbb{R}^{n \times p}$: discretized functional data.
- $u \in \mathbb{R}^n$: scores.

Smooth Functional PCA

- Problem setup [Huang et al., 2008]:

$$\min_{u,v} \|X - uv^T\|_F^2 + \alpha u^T u v^T \Omega v$$

- $X \in \mathbb{R}^{n \times p}$: discretized functional data.
- $u \in \mathbb{R}^n$: scores.
- $v \in \mathbb{R}^p$: loading function.

Smooth Functional PCA

- Problem setup [Huang et al., 2008]:

$$\min_{u,v} \|X - uv^T\|_F^2 + \alpha u^T u v^T \Omega v$$

- $X \in \mathbb{R}^{n \times p}$: discretized functional data.
- $u \in \mathbb{R}^n$: scores.
- $v \in \mathbb{R}^p$: loading function.
- Ω : roughness penalty matrix (e.g., integrated squared 2nd derivative).

Smooth Functional PCA

- Problem setup [Huang et al., 2008]:

$$\min_{u,v} \|X - uv^T\|_F^2 + \alpha u^T u v^T \Omega v$$

- $X \in \mathbb{R}^{n \times p}$: discretized functional data.
- $u \in \mathbb{R}^n$: scores.
- $v \in \mathbb{R}^p$: loading function.
- Ω : roughness penalty matrix (e.g., integrated squared 2nd derivative).
- α : tuning parameter

Smooth Functional PCA

- Problem setup [Huang et al., 2008]:

$$\min_{u,v} \|X - uv^T\|_F^2 + \alpha u^T u v^T \Omega v$$

- $X \in \mathbb{R}^{n \times p}$: discretized functional data.
- $u \in \mathbb{R}^n$: scores.
- $v \in \mathbb{R}^p$: loading function.
- Ω : roughness penalty matrix (e.g., integrated squared 2nd derivative).
- α : tuning parameter
- A power algorithm is defined to compute the PCs while incorporating smoothness penalty.

Smooth Functional PCA

- Problem setup [Huang et al., 2008]:

$$\min_{u,v} \|X - uv^\top\|_F^2 + \alpha u^\top u v^\top \Omega v$$

- $X \in \mathbb{R}^{n \times p}$: discretized functional data.
- $u \in \mathbb{R}^n$: scores.
- $v \in \mathbb{R}^p$: loading function.
- Ω : roughness penalty matrix (e.g., integrated squared 2nd derivative).
- α : tuning parameter
- A power algorithm is defined to compute the PCs while incorporating smoothness penalty.
- Consider the SVD of X as $X = UDV^\top$, where U and V have orthonormal columns and D is diagonal with ordered singular values. In particular, for $X = u_d v_d^\top$, v is the first principal component and $u = u_d$ gives the associated scores. With these representations, the power algorithm (described below) converges quickly, typically in only a few iterations.

Power Algorithm

Algorithm: Penalized Power Iteration

- 1 Initialize v .

Power Algorithm

Algorithm: Penalized Power Iteration

- 1 Initialize v .
- 2 Repeat until convergence:

Power Algorithm

Algorithm: Penalized Power Iteration

- 1 Initialize v .
- 2 Repeat until convergence:
 - 1 $u \leftarrow Xv$

Power Algorithm

Algorithm: Penalized Power Iteration

- 1 Initialize v .
- 2 Repeat until convergence:
 - 1 $u \leftarrow Xv$
 - 2 $v \leftarrow (I + \alpha\Omega)^{-1}X^\top u$

Power Algorithm

Algorithm: Penalized Power Iteration

- 1 Initialize v .
- 2 Repeat until convergence:
 - 1 $u \leftarrow Xv$
 - 2 $v \leftarrow (I + \alpha\Omega)^{-1}X^\top u$
 - 3 $v \leftarrow \frac{v}{\|v\|}$

Power Algorithm

Algorithm: Penalized Power Iteration

- 1 Initialize v .
- 2 Repeat until convergence:
 - 1 $u \leftarrow Xv$
 - 2 $v \leftarrow (I + \alpha\Omega)^{-1}X^\top u$
 - 3 $v \leftarrow \frac{v}{\|v\|}$
- 3 Update $X \leftarrow X - \sigma uv^\top$ and proceed to next component.

Power Algorithm

Algorithm: Penalized Power Iteration

- 1 Initialize v .
 - 2 Repeat until convergence:
 - 1 $u \leftarrow Xv$
 - 2 $v \leftarrow (I + \alpha\Omega)^{-1}X^\top u$
 - 3 $v \leftarrow \frac{v}{\|v\|}$
 - 3 Update $X \leftarrow X - \sigma uv^\top$ and proceed to next component.
- For notational convenience, we define $S(\alpha) = (I + \alpha\Omega)^{-1} \in \mathbb{R}^{m \times m}$, which simplifies expressions involving regularization. The penalty matrix Ω is set up so that larger values of the quadratic form $v^\top \Omega v$ mean rougher functions. This means that it penalizes functions that change quickly between time points.

Tuning Smoothness Parameters

- To select the optimal tuning parameters α efficiently, one can use a traditional Cross-Validation (CV) criterion and a computationally efficient closed-form Generalized Cross-Validation (GCV) criterion:

$$\text{CV}(\alpha) = \frac{1}{m} \sum_{j=1}^m \frac{\left[\{(I - S(\alpha))(X^T u)\}_{jj} \right]^2}{(1 - \{S(\alpha)\}_{jj})^2},$$

where $\{\cdot\}_{jj}$ denotes the j -th diagonal element.

$$\text{GCV}(\alpha) = \frac{1}{m} \frac{\|(I - S(\alpha))(X^T u)\|^2}{\left(1 - \frac{1}{m} \text{tr}\{S(\alpha)\}\right)^2}.$$

Sparse Functional PCA

- **Standard FPCA loadings are dense $\rightarrow$ hard to interpret.**
Sparsity highlights only the most relevant features, improving interpretability.

Sparse Functional PCA

- **Standard FPCA loadings are dense $\rightarrow$ hard to interpret.**
Sparsity highlights only the most relevant features, improving interpretability.
- **Dense loadings capture noise $\rightarrow$ unstable components.**
Sparsity filters out uninformative variation, yielding more robust PCs.

Sparse Functional PCA

- **Standard FPCA loadings are dense $\rightarrow$ hard to interpret.**
Sparsity highlights only the most relevant features, improving interpretability.
- **Dense loadings capture noise $\rightarrow$ unstable components.**
Sparsity filters out uninformative variation, yielding more robust PCs.
- **All grid points contribute equally $\rightarrow$ no feature selection.**
Sparsity acts as built-in feature selection, focusing on key time points or locations.

Sparse Functional PCA

- **Standard FPCA loadings are dense $\rightarrow$ hard to interpret.**
Sparsity highlights only the most relevant features, improving interpretability.
- **Dense loadings capture noise $\rightarrow$ unstable components.**
Sparsity filters out uninformative variation, yielding more robust PCs.
- **All grid points contribute equally $\rightarrow$ no feature selection.**
Sparsity acts as built-in feature selection, focusing on key time points or locations.
- Sparse FPCA formulation [Shen and Huang, 2008]:

$$\min_{u,v} \|X - uv^T\|_F^2 + p_\gamma(v)$$

where $p_\gamma(v)$ is a sparsity-inducing penalty.

Sparsity penalties

- **Lasso (Soft-thresholding):**

$$p_{\gamma}^{\text{soft}}(|\theta|) = 2\gamma|\theta|, \xrightarrow{\text{minimizer}} h_{\gamma}^{\text{soft}}(y) = \text{sign}(y)(|y| - \gamma)_+$$

- **Hard thresholding:**

$$p_{\gamma}^{\text{hard}}(|\theta|) = \gamma^2 I(|\theta| \neq 0), \xrightarrow{\text{minimizer}} h_{\gamma}^{\text{hard}}(y) = I(|y| > \gamma) y$$

- **SCAD penalty:**

$$p_{\gamma}^{\text{SCAD}}(|\theta|) = \begin{cases} 2\gamma|\theta|, & |\theta| \leq \gamma, \\ \frac{\theta^2 - 2a\gamma|\theta| + \gamma^2}{a-1}, & \gamma < |\theta| \leq a\gamma, \\ \frac{(a+1)\gamma^2}{2}, & |\theta| > a\gamma, \end{cases} \xrightarrow{\text{minimizer}} h_{\gamma}^{\text{SCAD}}(y) = \begin{cases} \text{sign}(y)(|y| - \gamma)_+, & |y| \leq 2\gamma, \\ \frac{(a-1)y - \text{sign}(y)a\gamma}{a-2}, & 2\gamma < |y| \leq a\gamma, \\ y, & |y| > a\gamma, \end{cases}$$

where $a = 3.7$ (Fan and Li [2001]).

sFPCA-rSVD Algorithm

Algorithm: sFPCA-rSVD

- 1 **Initialization:** Compute the best rank-one approximation of X using singular value decomposition (SVD), where $X \approx suv^\top$, and set $u \leftarrow su$.

sFPCA-rSVD Algorithm

Algorithm: sFPCA-rSVD

- ➊ **Initialization:** Compute the best rank-one approximation of X using singular value decomposition (SVD), where $X \approx suv^\top$, and set $u \leftarrow su$.
- ➋ **Iterate until convergence:**

sFPCA-rSVD Algorithm

Algorithm: sFPCA-rSVD

- ➊ **Initialization:** Compute the best rank-one approximation of X using singular value decomposition (SVD), where $X \approx suv^\top$, and set $u \leftarrow su$.
- ➋ **Iterate until convergence:**
 - ➊ **Update Left Singular Vector:** $u \leftarrow Xv$

sFPCA-rSVD Algorithm

Algorithm: sFPCA-rSVD

- ➊ **Initialization:** Compute the best rank-one approximation of X using singular value decomposition (SVD), where $X \approx suv^\top$, and set $u \leftarrow su$.
- ➋ **Iterate until convergence:**
 - ➊ **Update Left Singular Vector:** $u \leftarrow Xv$
 - ➋ **Update Right Singular Vector:** $v \leftarrow h_\gamma X^\top u$

sFPCA-rSVD Algorithm

Algorithm: sFPCA-rSVD

- ➊ **Initialization:** Compute the best rank-one approximation of X using singular value decomposition (SVD), where $X \approx suv^\top$, and set $u \leftarrow su$.
- ➋ **Iterate until convergence:**
 - ➊ **Update Left Singular Vector:** $u \leftarrow Xv$
 - ➋ **Update Right Singular Vector:** $v \leftarrow h_\gamma X^\top u$
 - ➌ **Normalize Right Singular Vector:** $v \leftarrow \frac{v}{\|v\|}$

Cross-Validation for Sparsity Selection

- **Sparsity parameter:** Tuning parameter controlling number of non-zero loadings in v ($0 =$ dense, $p =$ full sparsity).

Algorithm: K-fold CV for Sparsity Tuning

- 1 Randomly group the rows of side-by-side data matrix X into K roughly equal-sized groups, denoted as $X^1, \dots, X^K$.

Cross-Validation for Sparsity Selection

- **Sparsity parameter:** Tuning parameter controlling number of non-zero loadings in v ($0 =$ dense, $p =$ full sparsity).

Algorithm: K-fold CV for Sparsity Tuning

- 1 Randomly group the rows of side-by-side data matrix X into K roughly equal-sized groups, denoted as $X^1, \dots, X^K$.
- 2 For each sparse tuning parameter $j \in \{0, 1, \dots, p\}$ (level of sparsity), do the following:

Cross-Validation for Sparsity Selection

- **Sparsity parameter:** Tuning parameter controlling number of non-zero loadings in v ($0 =$ dense, $p =$ full sparsity).

Algorithm: K-fold CV for Sparsity Tuning

- 1 Randomly group the rows of side-by-side data matrix X into K roughly equal-sized groups, denoted as $X^1, \dots, X^K$.
- 2 For each sparse tuning parameter $j \in \{0, 1, \dots, p\}$ (level of sparsity), do the following:
 - 1 For $k = 1, \dots, K$, let X^{-k} be the data matrix X leaving out X^k . Apply Algorithm sFPCA-rSVD on X^{-k} and derive the FPC scores $u^{-k}(j)$. Then project X^k onto $u^{-k}(j)$ to obtain $v^k(j)$.

Cross-Validation for Sparsity Selection

- **Sparsity parameter:** Tuning parameter controlling number of non-zero loadings in v ($0 =$ dense, $p =$ full sparsity).

Algorithm: K-fold CV for Sparsity Tuning

- 1 Randomly group the rows of side-by-side data matrix X into K roughly equal-sized groups, denoted as $X^1, \dots, X^K$.
- 2 For each sparse tuning parameter $j \in \{0, 1, \dots, p\}$ (level of sparsity), do the following:
 - 1 For $k = 1, \dots, K$, let X^{-k} be the data matrix X leaving out X^k . Apply Algorithm sFPCA-rSVD on X^{-k} and derive the FPC scores $u^{-k}(j)$. Then project X^k onto $u^{-k}(j)$ to obtain $v^k(j)$.
 - 2 Calculate the K-fold CV scores defined as: (N is the number of grid points in X^k)

$$CV_j = \sum_{k=1}^K \frac{\|X^k - u^{-k}(j)v^k(j)\|^2}{N}$$

Cross-Validation for Sparsity Selection

- **Sparsity parameter:** Tuning parameter controlling number of non-zero loadings in v ($0 =$ dense, $p =$ full sparsity).

Algorithm: K-fold CV for Sparsity Tuning

- 1 Randomly group the rows of side-by-side data matrix X into K roughly equal-sized groups, denoted as $X^1, \dots, X^K$.
- 2 For each sparse tuning parameter $j \in \{0, 1, \dots, p\}$ (level of sparsity), do the following:
 - 1 For $k = 1, \dots, K$, let X^{-k} be the data matrix X leaving out X^k . Apply Algorithm sFPCA-rSVD on X^{-k} and derive the FPC scores $u^{-k}(j)$. Then project X^k onto $u^{-k}(j)$ to obtain $v^k(j)$.
 - 2 Calculate the K-fold CV scores defined as: (N is the number of grid points in X^k)

$$CV_j = \sum_{k=1}^K \frac{\|X^k - u^{-k}(j)v^k(j)\|^2}{N}$$

- 3 Select the degree of sparsity as $j_0 = \arg \min \{CV(j)\}$.

Comparison & Tuning

- **Smooth FPCA:**

Comparison & Tuning

- **Smooth FPCA:**

- Pros: Produces smooth eigenfunctions.

Comparison & Tuning

- **Smooth FPCA:**

- Pros: Produces smooth eigenfunctions.
- Algorithm: Penalized power iteration.

Comparison & Tuning

- **Smooth FPCA:**

- Pros: Produces smooth eigenfunctions.
- Algorithm: Penalized power iteration.
- Tuning: α (smoothness) via CV/GCV.

Comparison & Tuning

- **Smooth FPCA:**

- Pros: Produces smooth eigenfunctions.
- Algorithm: Penalized power iteration.
- Tuning: α (smoothness) via CV/GCV.

- **Sparse FPCA:**

Comparison & Tuning

- **Smooth FPCA:**

- Pros: Produces smooth eigenfunctions.
- Algorithm: Penalized power iteration.
- Tuning: α (smoothness) via CV/GCV.

- **Sparse FPCA:**

- Pros: Feature selection $\rightarrow$ interpretable.

Comparison & Tuning

- **Smooth FPCA:**

- Pros: Produces smooth eigenfunctions.
- Algorithm: Penalized power iteration.
- Tuning: α (smoothness) via CV/GCV.

- **Sparse FPCA:**

- Pros: Feature selection $\rightarrow$ interpretable.
- Algorithm: Iterative thresholding.

Comparison & Tuning

• Smooth FPCA:

- Pros: Produces smooth eigenfunctions.
- Algorithm: Penalized power iteration.
- Tuning: α (smoothness) via CV/GCV.

• Sparse FPCA:

- Pros: Feature selection $\rightarrow$ interpretable.
- Algorithm: Iterative thresholding.
- Tuning: γ (sparsity) via CV.

Comparison & Tuning

- **Smooth FPCA:**

- Pros: Produces smooth eigenfunctions.
- Algorithm: Penalized power iteration.
- Tuning: α (smoothness) via CV/GCV.

- **Sparse FPCA:**

- Pros: Feature selection $\rightarrow$ interpretable.
- Algorithm: Iterative thresholding.
- Tuning: γ (sparsity) via CV.

- **Combined Approaches:** Smooth + Sparse together.

$$\min_{u,v} \|X - uv^T\|_F^2 + \alpha v^T \Omega v + p_\gamma(v)$$

Comparison & Tuning

- **Smooth FPCA:**

- Pros: Produces smooth eigenfunctions.
- Algorithm: Penalized power iteration.
- Tuning: α (smoothness) via CV/GCV.

- **Sparse FPCA:**

- Pros: Feature selection $\rightarrow$ interpretable.
- Algorithm: Iterative thresholding.
- Tuning: γ (sparsity) via CV.

- **Combined Approaches:** Smooth + Sparse together.

$$\min_{u,v} \|X - uv^T\|_F^2 + \alpha v^T \Omega v + p_\gamma(v)$$

- Trade-off: **Variance explained vs Interpretability vs Smoothness.**

Regularized MFPCA

- **Context**

Regularized MFPCA

- **Context**

- Univariate FDA $\rightarrow$ one curve per subject

Regularized MFPCA

- **Context**

- Univariate FDA $\rightarrow$ one curve per subject
- **Multivariate FDA** $\rightarrow$ vector of curves per subject (EEG, growth, pollution)

Regularized MFPCA

- **Context**

- Univariate FDA $\rightarrow$ one curve per subject
- **Multivariate FDA** $\rightarrow$ vector of curves per subject (EEG, growth, pollution)
- MFPCA $\rightarrow$ **joint modes of variation** across functions

Regularized MFPCA

- **Context**

- Univariate FDA $\rightarrow$ one curve per subject
- **Multivariate FDA** $\rightarrow$ vector of curves per subject (EEG, growth, pollution)
- MFPCA $\rightarrow$ **joint modes of variation** across functions

- **Challenges**

Regularized MFPCA

- **Context**

- Univariate FDA $\rightarrow$ one curve per subject
- **Multivariate FDA** $\rightarrow$ vector of curves per subject (EEG, growth, pollution)
- MFPCA $\rightarrow$ **joint modes of variation** across functions

- **Challenges**

- Discretization & irregular grids $\rightarrow$ noise, missing data

Regularized MFPCA

- **Context**

- Univariate FDA $\rightarrow$ one curve per subject
- **Multivariate FDA** $\rightarrow$ vector of curves per subject (EEG, growth, pollution)
- MFPCA $\rightarrow$ **joint modes of variation** across functions

- **Challenges**

- Discretization & irregular grids $\rightarrow$ noise, missing data
- High dimensionality $\rightarrow$ unstable eigenfunctions

Regularized MFPCA

- **Context**

- Univariate FDA $\rightarrow$ one curve per subject
- **Multivariate FDA** $\rightarrow$ vector of curves per subject (EEG, growth, pollution)
- MFPCA $\rightarrow$ **joint modes of variation** across functions

- **Challenges**

- Discretization & irregular grids $\rightarrow$ noise, missing data
- High dimensionality $\rightarrow$ unstable eigenfunctions
- Cross-function correlation $\rightarrow$ requires joint smoothness

Regularized MFPCA

- **Context**

- Univariate FDA $\rightarrow$ one curve per subject
- **Multivariate FDA** $\rightarrow$ vector of curves per subject (EEG, growth, pollution)
- MFPCA $\rightarrow$ **joint modes of variation** across functions

- **Challenges**

- Discretization & irregular grids $\rightarrow$ noise, missing data
- High dimensionality $\rightarrow$ unstable eigenfunctions
- Cross-function correlation $\rightarrow$ requires joint smoothness

- **Proposed Solution: Penalized SVD**

Regularized MFPCA

- **Context**

- Univariate FDA $\rightarrow$ one curve per subject
- **Multivariate FDA** $\rightarrow$ vector of curves per subject (EEG, growth, pollution)
- MFPCA $\rightarrow$ **joint modes of variation** across functions

- **Challenges**

- Discretization & irregular grids $\rightarrow$ noise, missing data
- High dimensionality $\rightarrow$ unstable eigenfunctions
- Cross-function correlation $\rightarrow$ requires joint smoothness

- **Proposed Solution: Penalized SVD**

- **Smoothness penalties:** roughness on derivatives

Regularized MFPCA

- **Context**

- Univariate FDA $\rightarrow$ one curve per subject
- **Multivariate FDA** $\rightarrow$ vector of curves per subject (EEG, growth, pollution)
- MFPCA $\rightarrow$ **joint modes of variation** across functions

- **Challenges**

- Discretization & irregular grids $\rightarrow$ noise, missing data
- High dimensionality $\rightarrow$ unstable eigenfunctions
- Cross-function correlation $\rightarrow$ requires joint smoothness

- **Proposed Solution: Penalized SVD**

- **Smoothness penalties:** roughness on derivatives
- **Sparsity penalties:** ℓ_1 /SCAD shrinkage

Regularized MFPCA

- **Context**

- Univariate FDA $\rightarrow$ one curve per subject
- **Multivariate FDA** $\rightarrow$ vector of curves per subject (EEG, growth, pollution)
- MFPCA $\rightarrow$ **joint modes of variation** across functions

- **Challenges**

- Discretization & irregular grids $\rightarrow$ noise, missing data
- High dimensionality $\rightarrow$ unstable eigenfunctions
- Cross-function correlation $\rightarrow$ requires joint smoothness

- **Proposed Solution: Penalized SVD**

- **Smoothness penalties:** roughness on derivatives
- **Sparsity penalties:** ℓ_1 /SCAD shrinkage
- **Block-diagonal roughness matrix** for cross-function structure

Regularized MFPCA

- **Context**

- Univariate FDA $\rightarrow$ one curve per subject
- **Multivariate FDA** $\rightarrow$ vector of curves per subject (EEG, growth, pollution)
- MFPCA $\rightarrow$ **joint modes of variation** across functions

- **Challenges**

- Discretization & irregular grids $\rightarrow$ noise, missing data
- High dimensionality $\rightarrow$ unstable eigenfunctions
- Cross-function correlation $\rightarrow$ requires joint smoothness

- **Proposed Solution: Penalized SVD**

- **Smoothness penalties:** roughness on derivatives
- **Sparsity penalties:** ℓ_1 /SCAD shrinkage
- **Block-diagonal roughness matrix** for cross-function structure

- **Impact**

Regularized MFPCA

- **Context**

- Univariate FDA $\rightarrow$ one curve per subject
- **Multivariate FDA** $\rightarrow$ vector of curves per subject (EEG, growth, pollution)
- MFPCA $\rightarrow$ **joint modes of variation** across functions

- **Challenges**

- Discretization & irregular grids $\rightarrow$ noise, missing data
- High dimensionality $\rightarrow$ unstable eigenfunctions
- Cross-function correlation $\rightarrow$ requires joint smoothness

- **Proposed Solution: Penalized SVD**

- **Smoothness penalties:** roughness on derivatives
- **Sparsity penalties:** ℓ_1 /SCAD shrinkage
- **Block-diagonal roughness matrix** for cross-function structure

- **Impact**

- Produces **smooth, sparse, interpretable** joint modes

Regularized MFPCA

- **Context**

- Univariate FDA $\rightarrow$ one curve per subject
- **Multivariate FDA** $\rightarrow$ vector of curves per subject (EEG, growth, pollution)
- MFPCA $\rightarrow$ **joint modes of variation** across functions

- **Challenges**

- Discretization & irregular grids $\rightarrow$ noise, missing data
- High dimensionality $\rightarrow$ unstable eigenfunctions
- Cross-function correlation $\rightarrow$ requires joint smoothness

- **Proposed Solution: Penalized SVD**

- **Smoothness penalties:** roughness on derivatives
- **Sparsity penalties:** ℓ_1 /SCAD shrinkage
- **Block-diagonal roughness matrix** for cross-function structure

- **Impact**

- Produces **smooth, sparse, interpretable** joint modes
- More stable & applicable to high-dimensional multivariate FDA

Methodology: Multivariate Functional Data Framework

- A multivariate functional dataset is formed by **concatenating p functional data matrices**.

Methodology: Multivariate Functional Data Framework

- A multivariate functional dataset is formed by **concatenating p functional data matrices**.
 - Each variable: $X_i \in \mathbb{R}^{n \times m_i}$ where n : number of observations and m_i : grid points

Methodology: Multivariate Functional Data Framework

- A multivariate functional dataset is formed by **concatenating** p **functional data matrices**.
 - Each variable: $X_i \in \mathbb{R}^{n \times m_i}$ where n : number of observations and m_i : grid points
- **Rank-one approximation** (per variable):

$$X_i \approx u_i v_i^\top, \quad u_i \in \mathbb{R}^n, \quad v_i \in \mathbb{R}^{m_i}$$

Methodology: Multivariate Functional Data Framework

- A multivariate functional dataset is formed by **concatenating p functional data matrices**.
 - Each variable: $X_i \in \mathbb{R}^{n \times m_i}$ where n : number of observations and m_i : grid points
- **Rank-one approximation** (per variable):

$$X_i \approx u_i v_i^\top, \quad u_i \in \mathbb{R}^n, \quad v_i \in \mathbb{R}^{m_i}$$

- **Full data matrix:** $\mathbf{X} = [X_1 \ X_2 \ \cdots \ X_p] \in \mathbb{R}^{n \times \sum_{i=1}^p m_i}$

Methodology: Multivariate Functional Data Framework

- A multivariate functional dataset is formed by **concatenating** p **functional data matrices**.
 - Each variable: $X_i \in \mathbb{R}^{n \times m_i}$ where n : number of observations and m_i : grid points
- **Rank-one approximation** (per variable):

$$X_i \approx u_i v_i^\top, \quad u_i \in \mathbb{R}^n, \quad v_i \in \mathbb{R}^{m_i}$$

- **Full data matrix:** $\mathbf{X} = [X_1 \ X_2 \ \cdots \ X_p] \in \mathbb{R}^{n \times \sum_{i=1}^p m_i}$
- **Block structure example:**

$$X_i = \begin{bmatrix} x_{1i}(t_{i1}) & \cdots & x_{1i}(t_{i,m_i}) \\ \vdots & \ddots & \vdots \\ x_{ni}(t_{i1}) & \cdots & x_{ni}(t_{i,m_i}) \end{bmatrix}, \quad \mathbf{X} = [X_1 \ X_2 \ \cdots \ X_p]$$

Penalized Smooth MFPCA

- Standard FPCA loadings may be noisy; smoothness penalties (via block-diagonal Ω_i) improve structure and interpretability.

Penalized Smooth MFPCA

- Standard FPCA loadings may be noisy; smoothness penalties (via block-diagonal Ω_i) improve structure and interpretability.
- Let $\mathbf{X} \in \mathbb{R}^{n \times M}$ denote multivariate functional data, where $M = \sum_{i=1}^p m_i$. Its best rank-one approximation is $\mathbf{X} \approx uv^\top$, with $u \in \mathbb{R}^n$ (score vector) and $v \in \mathbb{R}^M$ (loading vector). A smoothness penalty is imposed on v .

Penalized Smooth MFPCA

- Standard FPCA loadings may be noisy; smoothness penalties (via block-diagonal Ω_i) improve structure and interpretability.
- Let $\mathbf{X} \in \mathbb{R}^{n \times M}$ denote multivariate functional data, where $M = \sum_{i=1}^p m_i$. Its best rank-one approximation is $\mathbf{X} \approx uv^\top$, with $u \in \mathbb{R}^n$ (score vector) and $v \in \mathbb{R}^M$ (loading vector). A smoothness penalty is imposed on v .
- The block-diagonal penalty matrix is $\mathbf{\Omega} = \text{blockdiag}(\Omega_1, \Omega_2, \dots, \Omega_p)$, where each $\Omega_i \in \mathbb{R}^{m_i \times m_i}$ is a univariate roughness penalty matrix.

Penalized Smooth MFPCA

- Standard FPCA loadings may be noisy; smoothness penalties (via block-diagonal Ω_i) improve structure and interpretability.
- Let $\mathbf{X} \in \mathbb{R}^{n \times M}$ denote multivariate functional data, where $M = \sum_{i=1}^p m_i$. Its best rank-one approximation is $\mathbf{X} \approx uv^\top$, with $u \in \mathbb{R}^n$ (score vector) and $v \in \mathbb{R}^M$ (loading vector). A smoothness penalty is imposed on v .
- The block-diagonal penalty matrix is $\mathbf{\Omega} = \text{blockdiag}(\Omega_1, \Omega_2, \dots, \Omega_p)$, where each $\Omega_i \in \mathbb{R}^{m_i \times m_i}$ is a univariate roughness penalty matrix.
- The penalized reconstruction error is

$$\min_{u,v} \|\mathbf{X} - uv^\top\|_F^2 + \alpha^\top (v^\top \mathbf{\Omega} v),$$

where $\alpha = (\alpha_1, \dots, \alpha_p)^\top$ controls smoothness.

MFPCA Power Algorithm

Algorithm: Regularized Power Iteration for Smooth MFPCA

- 1 Initialize v .

MFPCA Power Algorithm

Algorithm: Regularized Power Iteration for Smooth MFPCA

- 1 Initialize v .
- 2 Repeat until convergence:

MFPCA Power Algorithm

Algorithm: Regularized Power Iteration for Smooth MFPCA

- 1 Initialize v .
- 2 Repeat until convergence:
 - 1 $u \leftarrow \mathbf{X}v$

MFPCA Power Algorithm

Algorithm: Regularized Power Iteration for Smooth MFPCA

- ➊ Initialize v .
- ➋ Repeat until convergence:
 - ➊ $u \leftarrow \mathbf{X}v$
 - ➋ $v \leftarrow (I + \alpha\Omega)^{-1}\mathbf{X}^\top u$

MFPCA Power Algorithm

Algorithm: Regularized Power Iteration for Smooth MFPCA

- ➊ Initialize v .
- ➋ Repeat until convergence:
 - ➊ $u \leftarrow \mathbf{X}v$
 - ➋ $v \leftarrow (I + \alpha\Omega)^{-1}\mathbf{X}^\top u$
 - ➌ $v \leftarrow v/\|v\|$

MFPCA Power Algorithm

Algorithm: Regularized Power Iteration for Smooth MFPCA

- 1 Initialize v .
- 2 Repeat until convergence:
 - 1 $u \leftarrow \mathbf{X}v$
 - 2 $v \leftarrow (I + \alpha\Omega)^{-1}\mathbf{X}^\top u$
 - 3 $v \leftarrow v/\|v\|$
- 3 Update $\mathbf{X} \leftarrow \mathbf{X} - \sigma uv^\top$ to extract the next PC.

MFPCA Power Algorithm

Algorithm: Regularized Power Iteration for Smooth MFPCA

- ➊ Initialize v .
 - ➋ Repeat until convergence:
 - ➊ $u \leftarrow \mathbf{X}v$
 - ➋ $v \leftarrow (I + \alpha\Omega)^{-1}\mathbf{X}^\top u$
 - ➌ $v \leftarrow v/\|v\|$
 - ➌ Update $\mathbf{X} \leftarrow \mathbf{X} - \sigma uv^\top$ to extract the next PC.
- The smoothing operator is $\mathcal{S}(\alpha) = (I + \alpha\Omega)^{-1} \in \mathbb{R}^{M \times M}$.

MFPCA Power Algorithm

Algorithm: Regularized Power Iteration for Smooth MFPCA

- 1 Initialize v .
 - 2 Repeat until convergence:
 - 1 $u \leftarrow \mathbf{X}v$
 - 2 $v \leftarrow (I + \alpha\Omega)^{-1}\mathbf{X}^\top u$
 - 3 $v \leftarrow v/\|v\|$
 - 3 Update $\mathbf{X} \leftarrow \mathbf{X} - \sigma uv^\top$ to extract the next PC.
- The smoothing operator is $\mathbf{S}(\alpha) = (I + \alpha\Omega)^{-1} \in \mathbb{R}^{M \times M}$.
 - The smoothing parameter α is selected via generalized cross-validation (GCV), defined as

$$\text{GCV}(\alpha) = \frac{1}{M} \frac{\|(I - \mathbf{S}(\alpha))(\mathbf{X}^\top u)\|^2}{\left(1 - \frac{1}{M}\text{tr}\{\mathbf{S}(\alpha)\}\right)^2}.$$

Two-way Regularized MFPCA

Conclusion & Future Work

References I

- Jianqing Fan and Runze Li. Variable selection via nonconcave penalized likelihood and its oracle properties. *Journal of the American Statistical Association*, 96(456):1348–1360, 2001. doi: 10.1198/016214501753382273. URL <https://doi.org/10.1198/016214501753382273>.
- Hossein Haghbin, Yue Zhao, and Mehdi Maadooliat. Regularized multivariate functional principal component analysis, 2023. URL <https://arxiv.org/abs/2306.13980>.
- Clara Happ and Sonja Greven. Multivariate functional principal component analysis for data observed on different (dimensional) domains. *Journal of the American Statistical Association*, 113(522):649–659, February 2018. ISSN 1537-274X. doi: 10.1080/01621459.2016.1273115. URL <http://dx.doi.org/10.1080/01621459.2016.1273115>.
- Jianhua Z. Huang, Haipeng Shen, and Andreas Buja. Functional principal components analysis via penalized rank one approximation. *Electronic Journal of Statistics*, 2(none):678 – 695, 2008. doi: 10.1214/08-EJS218. URL <https://doi.org/10.1214/08-EJS218>.

References II

- Haipeng Shen Jianhua Z. Huang and Andreas Buja. The analysis of two-way functional data using two-way regularized singular value decompositions. *Journal of the American Statistical Association*, 104(488):1609–1620, 2009. doi: 10.1198/jasa.2009.tm08024. URL <https://doi.org/10.1198/jasa.2009.tm08024>.
- Yunlong Nie and Jiguo Cao. Sparse functional principal component analysis in a new regression framework. *Computational Statistics & Data Analysis*, 152:107016, 2020. ISSN 0167-9473. doi: <https://doi.org/10.1016/j.csda.2020.107016>. URL <https://www.sciencedirect.com/science/article/pii/S0167947320301079>.
- J. Ramsay and B.W. Silverman. *Functional Data Analysis*. Springer Series in Statistics. Springer, 2005. ISBN 9780387400808. URL https://books.google.com/books?id=mU3dop5wY_4C.

References III

Haipeng Shen and Jianhua Z. Huang. Sparse principal component analysis via regularized low rank matrix approximation. *Journal of Multivariate Analysis*, 99(6):1015–1034, 2008. ISSN 0047-259X. doi: <https://doi.org/10.1016/j.jmva.2007.06.007>. URL <https://www.sciencedirect.com/science/article/pii/S0047259X07000887>.

Bernard W Silverman. Smoothed functional principal components analysis by choice of norm. *The Annals of Statistics*, 24(1):1–24, 1996.